

HARIRAMAKRRISHNAN RAMACHANDRAN

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PROFESSIONAL EXPERIENCE

SOFTWARE ENGINEER – HCL Technologies | Chennai, India 09/2021 – 01/2025 (3.5 Years)

- Developed and maintained robust **Rust** backend services, focusing on **ownership**, **borrowing**, and **lifetime management** to ensure memory safety and prevent data races in high-concurrency scenarios.
- Authored comprehensive technical explanations and model solutions in **idiomatic Rust**, demonstrating precise reasoning and adherence to prompt requirements for AI code training datasets.
- Reviewed and evaluated AI-generated **Rust** code and reasoning, identifying subtle errors in algorithmic logic, correctness, and adherence to prompt specifications, contributing to model accuracy improvements of **15%**.
- Implemented asynchronous patterns using **Tokio** for I/O-bound operations, significantly improving the responsiveness and throughput of backend systems by **30%**.
- Designed and integrated solutions using **Serde** for efficient data serialization and deserialization, optimizing data transfer between microservices and reducing processing latency by **20%**.
- Contributed to the development of command-line interface (CLI) tools written in **Rust**, enhancing automation capabilities for data processing and system administration tasks.

SKILLS

Languages – Rust, Python, Java, SQL, R

Rust Ecosystem – Tokio, Actix, Serde, async Rust, ownership, borrowing, lifetimes, traits, generics, pattern matching, error handling

Backend Development – REST APIs, microservices, backend services, systems programming, CLI tools

AI & Data Training – AI code review, LLM evaluation, prompt engineering, data validation, model integrity testing, AI performance optimization

Software Engineering – Agile/Scrum, Git, CI/CD, Jenkins, Jira, Confluence, code reviews, incident response, debugging

Cloud & Databases – AWS (EC2, S3, Lambda, CloudWatch), Docker, PostgreSQL, MySQL, Oracle

EDUCATION AND TRAINING

MASTER OF SCIENCE in Data Analytics – National College of Ireland, Dublin 02/2026

BACHELOR OF ENGINEERING in Computer Science – SNS College of Technology, Coimbatore, India 01/2021

PROJECTS

AI Code Evaluation Platform

- Developed a backend service in **Rust** to ingest and process AI-generated code snippets and explanations for evaluation.
- Implemented validation logic to check for adherence to **Rust**'s memory safety rules and identify common algorithmic pitfalls.

- Designed REST APIs using **Actix** for seamless integration with a front-end interface for human reviewers, processing over **1000** evaluations weekly.

Asynchronous Data Processing Pipeline

- Engineered a high-throughput data pipeline using **Rust** and **Tokio** to process large volumes of training data, achieving a throughput of **50,000** records per second.
- Integrated **Serde** for efficient JSON parsing and serialization, reducing data handling overhead by **25%**.
- Built robust error handling mechanisms to ensure data integrity and provide detailed logging for troubleshooting across the asynchronous workflow.

ACHIEVEMENTS & CERTIFICATIONS

- IELTS Academic: Overall Band Score **7 out of 9**, issued by IELTS Official, Test Date: 28/AUG/2023
- **Full Stack: Java Spring Boot** and Angular (Great Learning)
- Certificate of Completion in Web Development from **University of California, Davis** (Coursera)
- Certificate of Completion in Python programming from **University of Michigan** (Coursera)